

Lagrange's Equation of Motion

Derivation from D'Alembert's Principle — arranged serially and clearly from the supplied handwritten derivation

- (1) Relation between the old Cartesian coordinates r_i and the new generalized coordinates $q_1, q_2, \dots, q_n$:

$$r_i = r_i(q_1, q_2, q_3, \dots, q_n)$$

- (2) For a virtual displacement, differentiating the above relation with respect to the generalized coordinates:

$$\delta r_i = (\partial r_i / \partial q_1) \delta q_1 + (\partial r_i / \partial q_2) \delta q_2 + \dots + (\partial r_i / \partial q_n) \delta q_n$$

$$\delta r_i = \sum_{j=1}^n (\partial r_i / \partial q_j) \delta q_j$$

- (3) D'Alembert's principle:

$$\sum_{i=1}^n (F_i - \dot{p}_i) \cdot \delta r_i = 0$$

$$\sum F_i \cdot \delta r_i = \sum \dot{p}_i \cdot \delta r_i$$

Left-hand side:

$$\begin{aligned} \sum_i F_i \cdot \delta r_i &= \sum_i F_i \cdot [\sum_j (\partial r_i / \partial q_j) \delta q_j] \\ &= \sum_j \sum_i [F_i \cdot (\partial r_i / \partial q_j)] \delta q_j \\ &= \sum_j Q_j \delta q_j \end{aligned}$$

where the generalized force is

$$Q_j = \sum_i F_i \cdot (\partial r_i / \partial q_j)$$

For a conservative system:

$$F_i = -\nabla_i V$$

$$Q_j = -\partial V / \partial q_j$$

Right-hand side:

$$\begin{aligned} \sum_i \dot{p}_i \cdot \delta r_i &= \sum_i (m_i \ddot{r}_i) \cdot [\sum_j (\partial r_i / \partial q_j) \delta q_j] \\ &= \sum_j \sum_i [m_i \ddot{r}_i \cdot (\partial r_i / \partial q_j)] \delta q_j \\ &= \sum_j \sum_i [d/dt (m_i \dot{r}_i \cdot \partial r_i / \partial q_j) - m_i \dot{r}_i \cdot d/dt (\partial r_i / \partial q_j)] \delta q_j \end{aligned}$$

Note: Using the product rule and the fact that m_i is constant,

$$\begin{aligned} d/dt (m_i \dot{r}_i \cdot \partial r_i / \partial q_j) &= m_i \ddot{r}_i \cdot \partial r_i / \partial q_j + m_i \dot{r}_i \cdot d/dt (\partial r_i / \partial q_j) \\ d/dt (\partial r_i / \partial q_j) &= \partial v_i / \partial q_j \end{aligned}$$

Therefore, the right-hand side becomes

$$\sum_j \sum_i [d/dt (m_i v_i \cdot \partial r_i / \partial q_j) - m_i v_i \cdot \partial v_i / \partial q_j] \delta q_j$$

Now introduce the total kinetic energy

$$T = \sum_i \frac{1}{2} m_i v_i \cdot v_i$$

$$\partial T / \partial q_j = \sum_i m_i v_i \cdot (\partial v_i / \partial q_j)$$

Also,

$$\partial T / \partial \dot{q}_j = \sum_i m_i v_i \cdot (\partial r_i / \partial q_j)$$

$$\sum_j [d/dt \{ \partial T / \partial \dot{q}_j \} - \partial T / \partial q_j] \delta q_j = 0$$

Putting the values of (5) and the above result into D'Alembert's equation (3):

$$\sum_j \{ Q_j - [d/dt (\partial T / \partial \dot{q}_j) - \partial T / \partial q_j] \} \delta q_j = 0$$

For a conservative system, $Q_j = -\partial V / \partial q_j$. Hence

$$\sum_j \{ -\partial V / \partial q_j - d/dt (\partial T / \partial \dot{q}_j) + \partial T / \partial q_j \} \delta q_j = 0$$

$$d/dt (\partial T / \partial \dot{q}_j) - \partial T / \partial q_j + \partial V / \partial q_j = 0$$

Rearranging:

$$d/dt (\partial T / \partial \dot{q}_j) - \partial / \partial q_j (T - V) = 0$$

Let us define the Lagrangian of the system:

$$L = T - V$$

Therefore, the equation of motion becomes

$$d/dt (\partial L / \partial \dot{q}_j) - \partial L / \partial q_j = 0$$

Euler-Lagrange Equation

$$d/dt (\partial L / \partial \dot{q}_j) - \partial L / \partial q_j = 0, \quad j = 1, 2, \dots, n$$

This derivation follows the supplied handwritten sequence, while typesetting the mathematical expressions clearly and correcting only obvious notation/formatting ambiguities.